

HEMAVAN TARNABY, Sweden

WMO: 021010

Lat: 65.808N

Lon: 15.085E

Elev: 458

StdP: 95.94

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -32.0	(c) -29.0	(d) -35.0	(e) 0.1	(f) -32.0	(g) -31.9	(h) 0.2	(i) -29.0	(j) 15.2	(k) -7.1	(l) 13.2	(m) -5.5	(n) 0.5	(o) 90	(p) 0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	22.9	15.3	20.7	14.2	18.6	13.3	16.4	20.9	15.0	19.2	13.9	17.7	3.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	10.8	18.7	13.1	9.9	17.0	11.9	9.2	15.7	47.6	20.9	43.5	19.2	40.3	17.8	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.3	8.7	7.3	-35.1	25.3	2.7	2.3	-37.1	26.9	-38.7	28.2	-40.2	29.5	-42.2	31.1
			-35.4	17.5	2.7	2.0	-37.3	18.9	-38.8	20.0	-40.3	21.1	-42.3	22.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	0.4	-10.1	-9.8	-6.6	-1.2	4.4	8.9	12.3	11.1	6.8	1.3	-4.6	-7.9
	DBStd	9.59	8.29	8.12	6.17	4.34	3.28	3.21	3.16	2.85	3.16	3.89	6.10	8.23
	HDD10.0	3666	622	556	515	337	176	61	13	21	104	269	437	554
	HDD18.3	6533	881	789	773	587	432	284	187	226	347	527	687	813
	CDD10.0	176	0	0	0	0	3	26	85	54	7	0	0	0
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH23.3	37	0	0	0	0	1	2	31	3	0	0	0	0
	CDH26.7	3	0	0	0	0	0	0	3	0	0	0	0	0
(14) Wind	WSAvg	2.6	2.8	2.8	2.8	2.6	2.8	2.8	2.5	2.3	2.5	2.5	2.4	2.6
(15) Precipitation	PrecAvg	721	69	52	51	30	34	49	85	73	72	71	65	70
	PrecMax	1054	235	195	161	65	97	97	173	130	158	192	152	238
	PrecMin	416	12	9	6	11	6	4	12	12	31	11	21	19
	PrecStd	154	48	39	40	12	19	26	41	31	28	37	37	45
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.0	5.2	11.1	20.4	23.2	26.2	23.5	17.7	11.4	7.0	4.6
		MCWB	2.7	2.7	2.1	5.9	11.6	14.2	18.0	15.7	12.2	8.7	5.9	3.4
	2%	DB	2.9	2.9	3.6	8.3	16.9	20.5	23.9	21.2	14.8	9.6	5.1	3.5
		MCWB	1.7	1.8	1.8	4.1	10.3	12.4	16.1	14.8	11.1	7.7	3.9	2.2
	5%	DB	1.5	1.7	2.5	6.1	13.9	18.0	21.6	19.2	13.2	8.1	3.3	2.3
		MCWB	0.4	0.6	0.9	3.1	8.7	11.5	15.2	14.2	10.5	6.4	2.1	1.1
	10%	DB	0.0	0.4	1.4	4.6	11.0	15.7	19.4	17.1	11.9	6.8	2.2	1.2
		MCWB	-1.0	-0.7	0.0	2.3	6.7	10.5	14.2	13.0	9.6	5.4	1.2	0.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	2.9	2.9	6.2	12.6	15.0	19.0	17.3	13.2	9.4	6.1	3.7
		MCDB	3.9	3.7	4.4	10.7	19.5	20.3	24.3	21.5	16.0	11.0	6.8	4.4
	2%	WB	1.7	1.9	2.0	4.6	10.6	13.3	17.3	15.8	11.8	7.7	3.9	2.4
		MCDB	2.6	2.8	3.2	7.6	16.0	18.4	22.0	19.7	14.0	9.2	5.0	3.2
	5%	WB	0.6	0.7	1.0	3.5	8.7	12.1	15.8	14.6	10.8	6.6	2.4	1.3
		MCDB	1.4	1.6	2.2	5.7	13.6	16.8	20.2	18.3	12.8	7.9	3.2	2.1
	10%	WB	-1.0	-0.6	0.1	2.5	7.0	11.0	14.6	13.4	9.8	5.4	1.3	0.2
		MCDB	0.0	0.3	1.2	4.4	10.6	15.4	18.7	16.6	11.7	6.7	2.1	1.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	8.4	9.6	8.4	8.1	8.3	9.5	9.1	7.5	5.1	5.4	7.0
		MCDBR	6.1	6.7	7.7	10.6	14.9	14.0	14.8	13.7	9.8	5.9	4.3	5.4
		MCWBR	5.6	6.1	6.2	7.0	8.9	7.0	7.8	8.0	6.4	4.7	3.8	4.9
	5% WB	MCDBR	6.3	6.6	6.8	8.6	14.4	12.1	12.9	12.2	8.5	5.9	4.2	5.6
		MCWBR	6.0	6.0	5.7	5.8	8.8	6.5	7.5	7.6	5.9	4.8	3.8	5.3
(40) Clear-Sky Solar Irradiance	taub		0.174	0.244	0.278	0.304	0.315	0.317	0.329	0.315	0.291	0.265	N/A	0.177
	taud		1.928	2.173	2.222	2.246	2.355	2.498	2.502	2.525	2.571	2.383	N/A	1.937
	Ebn at Noon		301	642	794	861	890	897	874	853	801	626	N/A	53
	Edh at Noon		18	52	81	102	102	92	89	78	60	42	N/A	5
(44) All-Sky Solar Radiation	RadAvg		0.09	0.51	1.87	3.73	4.79	4.85	4.53	3.40	1.83	0.67	0.16	0.02
	RadStd		0.01	0.11	0.24	0.30	0.33	0.45	0.56	0.37	0.22	0.09	0.02	0.00

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A
(47) Regional (34 neighbors)	DBAvg	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A

Nomenclature: See separate page